

## MATH 3310 HOMEWORK ASSIGNMENT 8

DUE ON THURSDAY 27 OCTOBER 2022

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

- (1) Prove the following statement:

*For all real numbers  $x$ ,  $y$ , and  $z$  one has  $x^2 + y^2 + z^2 \geq xy + yz + zx$ .*

- (2) Prove that the equation

$$x^4 + x^3 - 30 = 0$$

has a unique solution in the interval  $(2, 3)$ .

- (3) Give a proof of the following statement:

*For sets  $A$  and  $B$  one has  $A \cup B = A$  if and only if  $B \subseteq A$ .*

- (4) Prove that the following formula for sums of integers holds for all  $n \in \mathbb{N}$ :

$$\sum_{u=0}^n 2^u = 2^{n+1} - 1 .$$

- (5) Prove that no upper bound for the interval  $A = [0, 1)$  belongs to  $A$ .